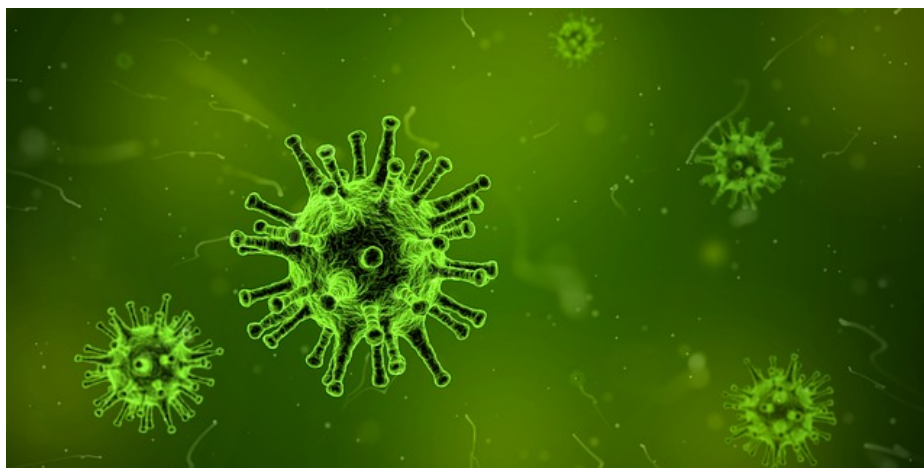


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Epidemic model

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Model for an epidemic

Simulate the spread of an epidemic.

Epidemic model

S stands for susceptible population, I is the already infected population, and R stands for the recovered population.

$$\begin{aligned}\frac{dS(t)}{dt} &= -\beta S(t)I(t) \\ \frac{dI(t)}{dt} &= \beta S(t)I(t) - \alpha I(t) \\ \frac{dR(t)}{dt} &= \alpha I(t)\end{aligned}$$

Tasks for the project in module 4

Finish the tasks for the SIR model that were optional in module 3; if you have not done them there, do them here. A repeat:

1. Implement Euler's method for the system, with $\alpha > 0$ and $\beta > 0$. The basic program for Euler's method has to be adapted to include three dependent variables, and three differential equations. Use the minimum of the two stepsizes determined in points 4 and 7 above.
2. Experiment with different initial conditions, and maybe with other combinations of α and β .

New tasks for module 4:

4. What combinations of (S, I, R) are equilibrium points? Is this what you expect?
5. Generate graphs of $S(t)$, $I(t)$ and $R(t)$ versus time. Is the sum of the populations still constant? Are these results reasonable for your epidemic?
6. Make a graph of solutions $I(t)$ versus time where the solutions are calculated with different sets of initial conditions. Do the same for $S(t)$ and $R(t)$ for those same sets of initial conditions. Can you make sense of the curves? What is the same for some curves, what differs?
7. You can also graph the solutions for several sets of initial conditions in phase space. The phase space is 3-dimensional, so you could try to plot 3-dimensional curves. Search for appropriate Python code for example with "matplotlib 3d parameter plot". Do you notice something special? You can also graph 2-dimensional projections of the solution curves: I versus S and R versus I . Include interesting graphs in your report.

Extending your model

After you have written and submitted your first report, in module 5 you have to go beyond the basic model. You should start thinking about what you would like to do then. What specific problem about an epidemic do you want to solve? That can be a pressing real-life problem, or you could investigate something you made up. What if ...?

So you could investigate what happens when there is some intervention: people are inoculated, the sick are better cared for and get better quicker. An intervention can be at $t = 0$, or you could simulate it to take effect after a certain time by including an if statement: If $t \geq 50$: and then your intervention happens, for example a change in the derivatives. Or maybe you want to include different effects at different times of the year. Then your final modelling problem could be What is the influence of ...?

You could also extend the model geographically, say there are two countries with some people travelling between the countries. Or you could consider a population that changes in size, because baby's are born, people immigrate or move away, people die from the illness or otherwise. How does this alter the spread of the disease?

You may be dissatisfied with this model, and you may alter the terms, such that they model your epidemic better. Then your modelling problem could be What is a mathematical model to predict ...? Your modelling problem is then a more theoretical problem, interesting for others working with the same kind of models.

Only in module 5 you will have to chose a way to extend your model. Finish your work on the basic model first and write and submit the first report before June 27th.

After the pages for the other problems, you will find a page for questions you might have about the project.

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